

Genetic report for longevity

Salt, Water and Electrolytes

Sensitivity to salt, sodium and water (osmolality), potassium, magnesium and calcium, salt and water intake, urinary excretion

Sérgio F · Report date: 02-10-2026

The kidney maintains sodium, potassium, magnesium and water within narrow limits, but with age the thirst decreases, the ability to concentrate urine falls and the sensitivity to salt increases. This report reads the genes of salt sensitivity (AGT, ADD1, GNB3, UMOD, CYP11B2, NEDD4L), the loci of sodium and water, potassium, magnesium and calcium in blood, and polygenic scores of salt and water intake and urinary excretion of sodium and potassium. It is useful because three drugs that already take touch these balances: dapagliflozin (loss of sodium and water), telmisartan (potassium retardation) and lithium (dependence on sodium and water).

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

16

variants

12

in the chip

2

imputed
($DR2 \geq 0.8$)

2

not found

10

Scores with
percentile

Overall reading. Salt and electrolytes with typical genetics, with two points to monitor: magnesium and potassium from the diet. In sodium: AGT M235T, GNB3 C825T and ADD1 Gly460Trp heterozygotes (the latter against); UMOD with 1 copy of the allele that lowers uromodulin, in favor. The Health Report puts the sensitivity to salt at the 49th percentile. In the behavior: salt added at the 68th percentile (PGS) and 77 in the loci score, against; water drink per day in the 56 and 71, informative. In urine: sodium at the 37th percentile and potassium in the 5; potassium in urine reflects mainly vegetables, fruit and legumes, and this is the clearest sign against. In the blood: low magnesium score in the 90 and MUC1 C/C percentile (2 copies of the allele that lowers magnesium), against; potassium in the 83 and sodium in the 85, informative; calcium in the 22 and CASR Ala986/Ala986, without judgment. With telmisartan (which rises potassium) and dapagliflozin (which removes sodium and water through urine), the most important is hydration and the origin of potassium.

Favourable

- **UMOD rs4293393 GA:** UMOD: the T allele of the gene (A in strand read) increases uromodulin, which activates the cotransporter NKCC2; more sodium retention (and more protection against urinary infections and calculations).

Unfavourable

- **Potassium in urine (21 512 variants):** the 5th percentile
- **Low magnesium in blood:** the 90th percentile
- **Salt added and taste for salted (GWAS loci):** the 77th percentile
- **ADD1 rs4961 GT:** α -aducin Gly460Trp: the Trp allele increases sodium resorption in the tubule; more sensitivity to salt.
- **MUC1 rs4072037 CC:** MUC1: locus of magnesium in blood.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 12 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Sodium and salt sensitivity

The renin-angiotensin-aldosterone system (AGT, CYP11B2), the transport of sodium in the kidney (ADD1, NEDD4L, UMOD) and the protein G (GNB3) explain part of the sensitivity to salt. The effects are small and vary between studies.

Ageing mechanisms: Vascular Aging; Arterial stiffness

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Salt sensitivity (5 variant score)	Typical / intermediate	the 49th percentile	Typical labeling. Source: Health Report.

Favourable

- UMOD rs4293393 GA

Unfavourable

- ADD1 rs4961 GT

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
UMOD	rs4293393 16:20353266 · A/G	GA	chip	A (1; 80%)	moderate	UMOD: the T allele of the gene (A in strand read) increases uromodulin, which activates the cotransporter NKCC2; more sodium retention (and more protection against urinary infections and calculations). 1 copy of the less common allele, bound to less uromodulin and less sensitivity to salt.

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADD1	rs4961 4:2904980 · G/A/T	GT	chip	T (1; 20%)	moderate	α-aducin Gly460Trp: the Trp allele increases sodium resorption in the tubule; more sensitivity to salt. G/T = Gly460Trp heterozygous.

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
AGT	rs699 1:230710048 · A/G	AG	chip	G (1; 41%)	moderate	AGT M235T: the Thr235 allele (G in the strand read, gene in reverse strand) increases angiotensinogen and salt sensitivity. Gene in reverse strand: the allele T235 is G in the strand read. A/G = M235T heterozygous.
GNB3	rs5443 12:6845711 · C/T	CT	chip	T (1; 31%)	weak	GNB3 C825T: T allele was associated with more sensitivity to salt and more response to diuretics.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP11B2	rs1799998 8:142918184 · A/G/T	AA	chip	—	weak	Aldosterone synthase -344C>T: small and inconsistent associations with aldosterone. Information.
NEDD4L		AG	chip	—	weak	

	rs4149601 18:58149559 · G/A					NEDD4L: regulates the sodium channel ENAC; inconsistent associations. Information.
ACE (I/D)	rs4343 17:63488670 · G/A	AG	chip	—	weak	Tag of the ACE I/D polymorphism (the G allele marks D, with more activity). Information.

Water, osmolality and ingestion

Plasma osmolality is regulated by vasopressin and thirst; sodium loci in blood and drinking water per day are few and have a small effect.

Ageing mechanisms: Metabolic Homeostasis Decline

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Drinking water per day (4993 variants)	No judgement (informative)	the 56th percentile · z=0.17	Higher score = more water cups. Information. PGS001069 (PGS Catalog; snpnet; Water intake; PGP000244, PMID 35324888). Variants with genotype: 4389 of 4993 (covering of 89.4% coefficients).
Sodium in blood	No judgement (informative)	the 85th percentile · z=1.04	Higher score = higher sodium. Information. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 74 of 82 loci with genotype. Studies: Choe EK (Sodium levels); Verma A (mean, inv-norm transformed)). Variants with genotype: 74 of 82 (covering of 54.9% coefficients).
Drinking water per day (loci of GWAS)	No judgement (informative)	the 71st percentile · z=0.55	Higher score = more water. Information. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 24 of 29 loci with genotype. Studies: Cole JB (Water consumption) (UKB data field 1528). Variants with genotype: 24 of 29 (covering of 81.7% coefficients).

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NFAT5	rs17298067 16:69667584 · G/C	GG	imputation DR2=1.00	—	moderate	NFAT5: response of cells to osmolality. Information: G/G = 2 copies of the sodium allele slightly lower in blood. Meaning: GWAS Catalog: Sodium (mean, inv-norm transformed) (Verma A 2024; p=1e-106); G ↓ allele
SLC4A10	rs62188151 2:161453338 · C/A	CA	imputation DR2=1.00	—	moderate	SLC4A10: bicarbonate transporter; sodium in blood. Information: C/A = 1 copy of the slightly lower sodium bound allele (C) and 1 of the least common allele (A). Meaning: GWAS Catalog: Sodium (mean, inv-norm transformed) (Verma A 2024; p=4e-37); C ↓ allele

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FGF21	— — · —	—	SNP not found in the data	—	moderate	FGF21: hormone that increases the thirst and water intake. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Potassium, magnesium and calcium

The magnesium in the blood depends on TRPM6 and CNNM2 (kidney and intestine), MUC1 and SHROOM3; the calcium of CASR. Potassium depends mainly on the diet and drugs.

Ageing mechanisms: Mitochondrial dysfunction; Metabolic Homeostasis Decline

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Calcium in blood (48 395 variants)	No judgement (informative)	the 22nd percentile · z=-0.77	Information. PGS001893 (PGS Catalog; Penalized regression (bigstatsr); Calcium; PGP000263, PMID 34995502). Variants with genotype: 46656 of 48395 (covering of 96.5% coefficients).
Low magnesium in blood	Unfavourable	the 90th percentile · z=1.26	Higher score = lower magnesium. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 36 of 40 loci with genotype. Studies: Verma A (magnesium (Mg, mean, inv-norm transformed)). Variants with genotype: 36 of 40 (covering of 89.8% coefficients).
Potassium in blood	No judgement (informative)	the 83rd percentile · z=0.99	Higher score = potassium. Information. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 96 of 107 loci with genotype. Studies: Verma A (potassium (mean, inv-norm transformed). Variants with genotype: 96 of 107 (covering of 70.8% coefficients).

Favourable

None.

Unfavourable

- Low blood magnesium (the 90th percentile)
- MUC1 rs4072037 CC

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MUC1	rs4072037 1:155192276 · C/A/G/T	CC	chip	C (2; 45%)	moderate	MUC1: locus of magnesium in blood. Meaning: GWAS Catalog: Magnesium levels (Meyer TE 2010; p=2e-36); C ↓ allele

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SHROOM3	rs13146355 4:76490987 · G/A	AG	chip	G (1; 57%)	moderate	SHROOM3: blood magnesium locus and renal function. Meaning: GWAS Catalog: magnesium (Mg, mean, inv-norm transformed) (Verma A 2024; p=4e-59); G ↓ allele

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CASR	rs1801725 3:122284910 · G/A/T	GG	chip	—	moderate	CASR A986S: T allele increases calcium in the blood. Information: G/G = Ala986/Ala986, without the Allele Being that rises calcium in the blood. Meaning: GWAS Catalog: Calcium levels (Sinnott-Armstrong N 2021; p=0e+00); allele T ↑
TRPM6		CC	chip	—	moderate	TRPM6: magnesium channel of the kidney and intestine.

rs3750425
9:74762494 ·
C/A/T

CNNM2	rs3740393 10:102876898 · G/C/T	GG	chip	—	moderate	CNNM2: magnesium transporter.
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SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
KCNJ1	— — · —	—	SNP not found in the data	—	moderate	KCNJ1 (ROMK): renal potassium channel. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Salt intake and urinary excretion

Sodium and potassium in urine mainly reflect the diet; scores measure the genetic trend of behavior (salt added, drinking water) and excretion.

Ageing mechanisms: Vascular Aging; Nutrient Sensing Deregulation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Salt added to food (6651 variants)	Typical / intermediate	the 68th percentile · z=0.50	Higher score = more salt added. PGS001034 (PGS Catalog; snpnet; Salt added to food; PGP000244, PMID 35324888). Variants with genotype: 5934 of 6651 (covering of 90.3% coefficients).
Sodium in urine (37 568 variants)	Typical / intermediate	the 37th percentile · z=-0.34	Higher score = more sodium in urine (reflects the intake). PGS002007 (PGS Catalog; Penalized regression (bigstatsr); Sodium in urine; PGP000263, PMID 34995502). Variants with genotype: 36375 of 37568 (covering of 96.9% coefficients).
Potassium in urine (21 512 variants)	Unfavourable	the 5th percentile · z=-1.64	Higher score = more potassium in urine (reflection plus intake of vegetables and fruit). PGS001974 (PGS Catalog; Penalized regression (bigstatsr); Potassium in urine; PGP000263, PMID 34995502). Variants with genotype: 20853 of 21512 (covering of 96.9% coefficients).
Salt added and taste for salted (GWAS loci)	Unfavourable	the 77th percentile · z=0.73	Higher score = more salt. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 37 of 43 loci with genotype. Studies: Cole JB (Added salt consumption); May-Wilson S (Salty food like). Variants with genotype: 37 of 43 (covering of 87.9% coefficients).

Favourable

None.

Unfavourable

- Potassium in urine (21 512 variants) (the 5th percentile)
- Salt added and taste for salted (GWAS loci) (the 77th percentile)

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Vascular Aging	ADD1 GT; Potassium in urine (21 512 variants) (the 5th percentile); Salt added and taste for salted (GWAS loci) (the 77th percentile)	UMOD GA	Salt sensitivity (5 variant score) (the 49th percentile); AGT AG; Salt added to food (6651 variants) (the 68th percentile); Sodium in urine (37 568 variants) (the 37th percentile)
Metabolic Homeostasis Decline	Low blood magnesium (the 90th percentile); MUC1 CC	—	SHROOM3 AG
Nutrient Sensing Deregulation	Potassium in urine (21 512 variants) (the 5th percentile); Salt added and taste for salted (GWAS loci) (the 77th percentile)	—	Salt added to food (6651 variants) (the 68th percentile); Sodium in urine (37 568 variants) (the 37th percentile)
Mitochondrial dysfunction	Low blood magnesium (the 90th percentile); MUC1 CC	—	SHROOM3 AG
Arterial stiffness	ADD1 GT	UMOD GA	Salt sensitivity (5 variant score) (the 49th percentile); AGT AG

Practical part

Ideas for salt, water and electrolytes, crossed with the genetic plan v2 and current medication (isotretinoin, retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin, lithium at low dose). The current stack remains as it is; everything is suggestion (not applied). See also the Rhym and Uric Acid report for kidney stones.

1. Potassium of food, not supplements

Genetics: Potassium in urine p5; potassium in blood p83

More vegetables, fruit, legumes, potato and dried fruits. With telmisartan and a high blood potassium score, no potassium supplements or salt with potassium chloride without blood potassium analysis.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

2. Moderate salt, without salt at the table

Genetics: Salt added p68 and p77; ADD1 Gly460Trp

Cooking with little salt and not joining the table; herbs, lemons and spices. Do not cut the salt at zero on days of very sweat (dapagliflozin and training).

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. Regular hydration, more in days of heat and training

Genetics: Water drink p56 (informative); NFAT5 and SLC4A10 informative

Dapagliflozin removes water and sodium through urine and lithium rises when there is dehydration. Drinking throughout the day (light urine), plus 0.5–1 L per hour of sweat training, with sodium in long sessions.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

4. Days of vomiting or diarrhea

Genetics: —

With retatrutide, vomiting may occur; in these days sodium, potassium and water are lost. Restoring with oral serum and talking to the doctor about paused dapagliflozin while lasting (rule of days of disease).

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

5. Maintaining magnesium

Genetics: Low magnesium p90 score; MUC1 C/C

Magnesium glycinate already taken makes sense with this genetics. Add magnesium in the annual blood analysis.

V2 genetic plan and medication: Already in the plan. • *suggestion (not applied)*

Avoiding or with caution

- Salt with potassium chloride or potassium supplements without analysis (telmisartan). *suggestion (not applied)*
- NSAID for days followed with telmisartan and dapagliflozin: the combination overloads the kidney. *suggestion (not applied)*
- Long Sauna or heat training without replenishing water and sodium (dapagliflozin, lithium). *suggestion (not applied)*
- Sugared isotonic beverages in daily life. *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. The I/D polymorphism of ACE is an indel read by tag; rare tubules (Gitelman, Bartter, Liddle) are not read.
- The salt sensitivity genes have small and often unreplicated effects; the diet, kidneys and drugs weigh much more.
- The electrolytes in the blood today depend mainly on medication (dapagliflozin, telmisartan, lithium) and hydration.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001034 (PGS Catalog; snpnet; Salt added to food; PGP000244, PMID 35324888 · PGS001069 (PGS Catalog; snpnet; Water intake; PGP000244, PMID 35324888 · PGS002007 (PGS Catalog; Penalized regression (bigstatsr); Sodium in urine; PGP000263, PMID 34995502 · PGS001974 (PGS Catalog; Penalized regression (bigstatsr); Potassium in urine; PGP000263, PMID 34995502 · PGS001893 (PGS Catalog; Penalized regression (bigstatsr); Calcium; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Jeunemaitre X et al. Cell 1992 (angiotensinogen, AGT).
- Cusi D et al. Lancet 1997 (alpha-aducine and salt sensitivity).
- Trudu M et al. Common noncoding UMOD gene variants induce salt sensitivity and kidney damage by increasing uromodulin expression. Nat Med 2013.
- Siffert W et al. Nat Genet 1998 (GNB3 C825T variant).
- Meyer TE et al. Genome-wide association studies of serum magnesium, potassium, and sodium concentrations identify six loci influencing serum magnesium levels. PLoS Genet 2010.
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- O'Seaghdha CM et al. Meta-analysis of genome-wide association studies identifies six new loci for serum calcium concentrations. PLoS Genet 2013.
- Zanetti D et al. Identification of 22 novel loci associated with urine biomarkers of albumin, sodium, and potassium excretion. Kidney Int 2019.
- Tanigawa Y et al. PLoS Genet 2022 (PGS001034, PGS001069); Primé F et al. Am J Hum Genet 2022 (PGS001893, PGS001974, PGS002007).
- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

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